PXL – Digital
421280 Software Analysis

Documentation of Requirements using NL

Week 08 – period 01

Luc Doumen

Nathalie Fuchs

DE HOGESCHOOL MET HET NETWERK

Elfde-Liniestraat 24, 3500 Hasselt, www.pxl.be

Content

- Natural language: what, language effects, etc.
- Five most relevant “transformations”
- Requirements writing according to the template
- – Dealing with transformations
- Recap IEEE 830 – SRS, standardized document structures & requirements classification
- Fit criteria
- Requirement card & attributes
- Key learning points
- Questions & answers

Natural language

Natural Language (1)

- Every stakeholder can “easy” understand it
- Is universal, can be used to describe the desired circumstances

- *If we could look into each others brains, we wouldn’t need documentation…*
  - *Words can help us to communicate*

Natural Language (2)

- … but is often **ambiguous** and **interpretable**

- Examples in English
  - The chicken is ready to eat
  - They are hunting dogs

Natural Language (2)

- … but is often **ambiguous** and **interpretable**

- Examples in English
  - The chicken is ready to eat
  - They are hunting dogs

Natural Language (2)

- … but is often **ambiguous** and **interpretable**

- Examples in Dutch
  - Wie wast Tuur?
  - Ik leer kinderen tekenen

Natural Language (2)

- … but is often **ambiguous** and **interpretable**

- Examples in Dutch
  - Wie wast Tuur?
  - Ik leer kinderen tekenen

Language Effects

- It is about perceptions and transformations of the documentation of the perceived 'things' ➔ SUBJECTIVE
- 'What is said' versus 'What is meant'
- ***It is the responsibility of the requirement engineer to determine what the originator of the requirement really meant!***
- ***So how to exploit transformational effects?***

- Five most relevant transformations
  - Nominalization - Nouns without reference index
  - Universal quantifiers - Incompletely specified conditions
  - Incompletely specified process words

Five most relevant “transformations”

Five most relevant Transformations (1)

1. Nominalization
   - Using nouns for complex processes
   - E.g. concepts like ‘registration’, ‘administration’, ‘acquisition’, etc.
   - Only to be used when the process is completely clear / defined ➔ Glossary!

Five most relevant Transformations (1)

1. Nominalization - example
   - After a **system breakdown** a **restart** shall automatically be initiated
   - **Terms** must be analyzed more precisely
     - What failures and exceptions might lead to a system breakdown?
     - What will happen during a restart?
     - When will this restart be executed?
     - Who initiates the restart?
     - What failure and exceptions might happen during the restart?

- **Is this clear for every stakeholder?**
- **What is a system breakdown?**
- **Apparently, I can restart?**

Five most relevant Transformations (2)

2. Nouns without reference index
   - E.g. “The data will be shown on the terminal to the user”
     - What data, on which terminal, to which user?
     - Better: “The system shall show the billing data on the terminal where the registered user is signed on”

3. Universal quantifiers
   - E.g. ‘never’, ‘always’, ‘none’, ‘all’, ‘everybody’, etc.

4. Incompletely specified conditions

   E.g. ‘in case’, ‘when’, ‘dependent on’, ...

Five most relevant Transformations (3)

5. Incompletely specified process words
   - E.g. ‘to transfer’: from where, where to?
   - Passive phrasing ➔ active phrasing

Fuzzy terms list

Mostly	As needed
Might	Make sense
Appropriate	Might make sense
Graceful	At minimum
Major	Slowly
May be of use	Including but not limited to
And/or	Suitable
Various	Clean and stable interface
Several	

Exercise – new clock radio (part 01)

- As a team determine what are the defects in the (natural language) requirements for the new clock radio? (cf. next slide)

- It is a defect when it is a violation of a rule!

Exercise – transformations (continued)

Topic – New clock radio

The development of the innovative alarm clock and music system shall raise the profit of the company by 10% within 3 months after delivery to the wholesale dealer.

The clock shall produce no tic sound. Because people don't like to be woken up abruptly, the new clock radio shall wake up the sleeping people more smoothly than any other alarm clock.

The luminosity of the lighted digits shall be changeable continuously.

The clock radio shall be small and lightweight, because the users also want to use it on travel. Therefore the clock radio shall be usable with different power voltages. To achieve a minimum weight the alarm clock must not contain any iron parts.

The alarm clock shall be able to wake its environment with music or a wake-up call.

The clock shall be very easy to use. The snooze key shall be easily accessible. The clock shall withstand a heavy blow from a sleeper and a fall from 50 cm.

The clock shall be very low in energy consumption. The system may only be operated with batteries of type 3434.

Exercise – transformations (continued)

Topic – New clock radio

1 The development of the innovative alarm clock and music system shall raise the profit of the company by 10% within 3 months after delivery to the wholesale dealer. 3

2 The clock shall produce no tic sound. Because people don't like to be woken up abruptly, the new clock radio shall wake up the sleeping people more smoothly than any other alarm clock.

4 The luminosity of the lighted digits shall be changeable continuously.

5 The clock radio shall be small and lightweight, because the users also want to use it on travel. Therefore the clock radio shall be usable with different power voltages. To achieve a minimum weight the alarm clock must not contain any iron parts. 7 6

8 The alarm clock shall be able to wake its environment with music or a wake-up call.

9 The clock shall be very easy to use. The snooze key shall be easily accessible. 10

11 The clock shall withstand a heavy blow from a sleeper and a fall from 50 cm.

12 The clock shall be very low in energy consumption. The system may only be operated with batteries of type 3434. 13

Requirements writing according to the template

Dealing with transformations

- Generic, syntactical requirements templates
  - The blueprint that determines the syntactical structure of a single SYSTEM <u>REQUIREMENT</u>
  - For complete and unambiguous sentences
  - To create precisely phrased requirements

- ***<u>Five</u>*** steps to write requirements according to the template

Dealing with transformations

1. Determine legal obligation/relevance
   - Fixing of liability
   - Strongly recommended (should)
   - Used in the future (will)
   - E.g., ‘shall’, ‘should’& ‘will’
   - If liabilities change, then requirements change too
   - Use of attributes is another way to document liabilities

Dealing with transformations

1. Determine legal obligation/relevance
   ➔ verb convention

Mate van verplichting	Hulpwerkwoord in het Nederlands	Hulpwerkwoord in het Engels
Wettelijk verplicht	Moeten	Shall
Noodzakelijk	Kunnen	Should
Wenselijk	Zullen	Will

Dealing with transformations

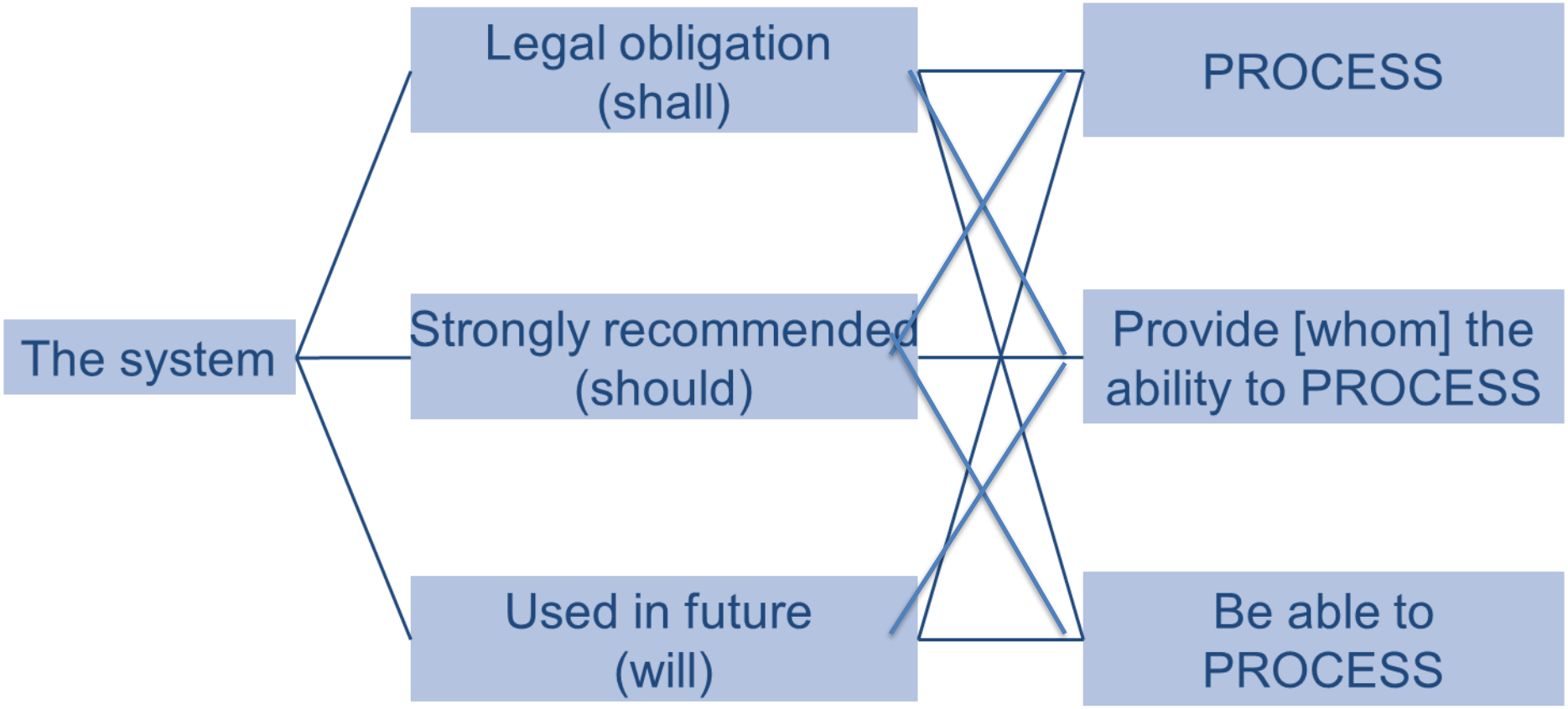
2. Determine the core of the requirement
   - The core of the system requirement is the required functionality (e.g., print, calculate, transfer)
   - Processes are activities or events: they shall be described using verbs
   - It might be necessary to add the (process) verbs to the glossary

Dealing with transformations

3. Characterize the activity
   - The way the system works is closely linked to the process words
   - Three types of system activities
     - *Autonomous system activity*: the system carries out the process by itself
     - *User interaction*: the system provides the user with process functionality
     - *Interface requirement*: the system carries out the process dependent on a third factor (for instance a different system), remains passive and waits for an external event

Dealing with transformations

3. Characterize the activity - basic structure



Dealing with transformations

- The first three steps …
  - Our planned system should offer a print option
    - Option is indispensable for us
    - “Print” is the process word
    - The system gives the administrator the option to
  - Requirement no. 1, version 1:
    - The system **shall** provide the **administrator** the ability to **print**

Dealing with transformations

4. Insert objects
   - Further elements are required for the completion
   - A closer or supplementing characterization of the process word
   - ‘What’, ‘where’

   - Requirement no. 1, version 2:
     - The system **shall** provide the **administrator** the ability to **print** the error message to the network printer

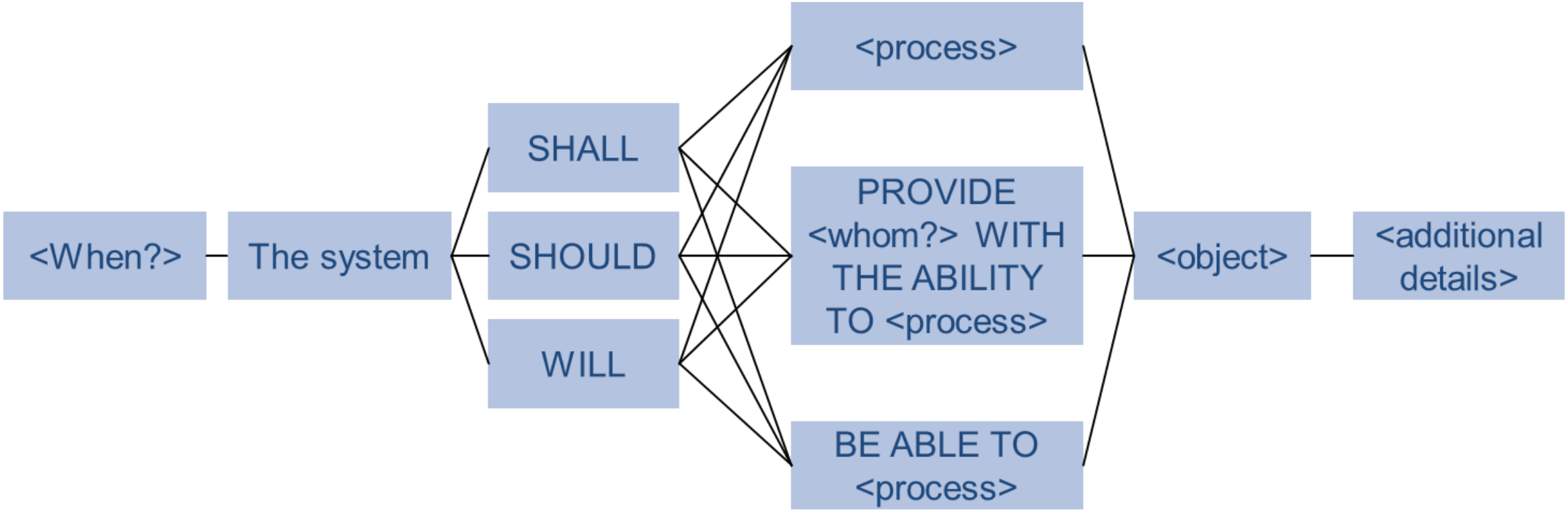
Dealing with transformations

5. Determine conditions
   - The functionality is only given or provided under certain logical or temporal conditions

   - Requirement no. 1, version 3:
     - If an error message has been generated, the system **shall** provide the **administrator** the ability to **print** the error message to the network printer

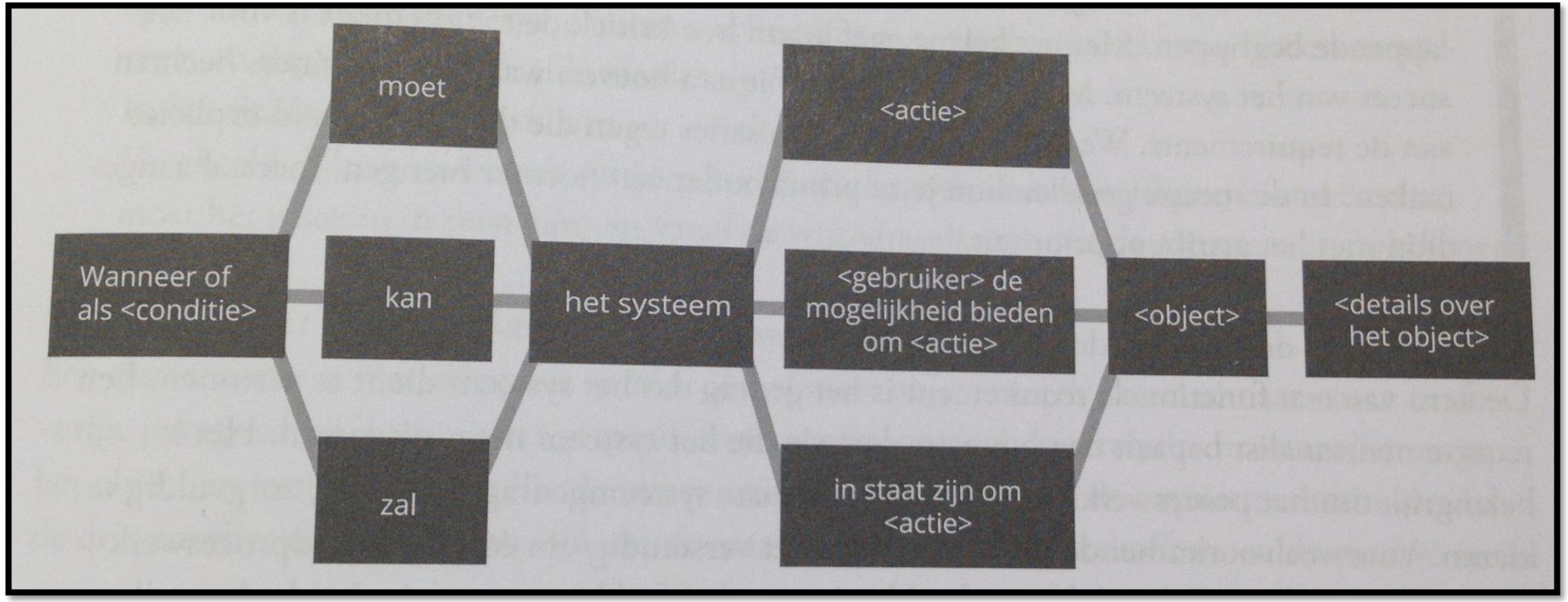
Dealing with transformations

Complete requirement template (according to Rupp – English version)



Dealing with transformations

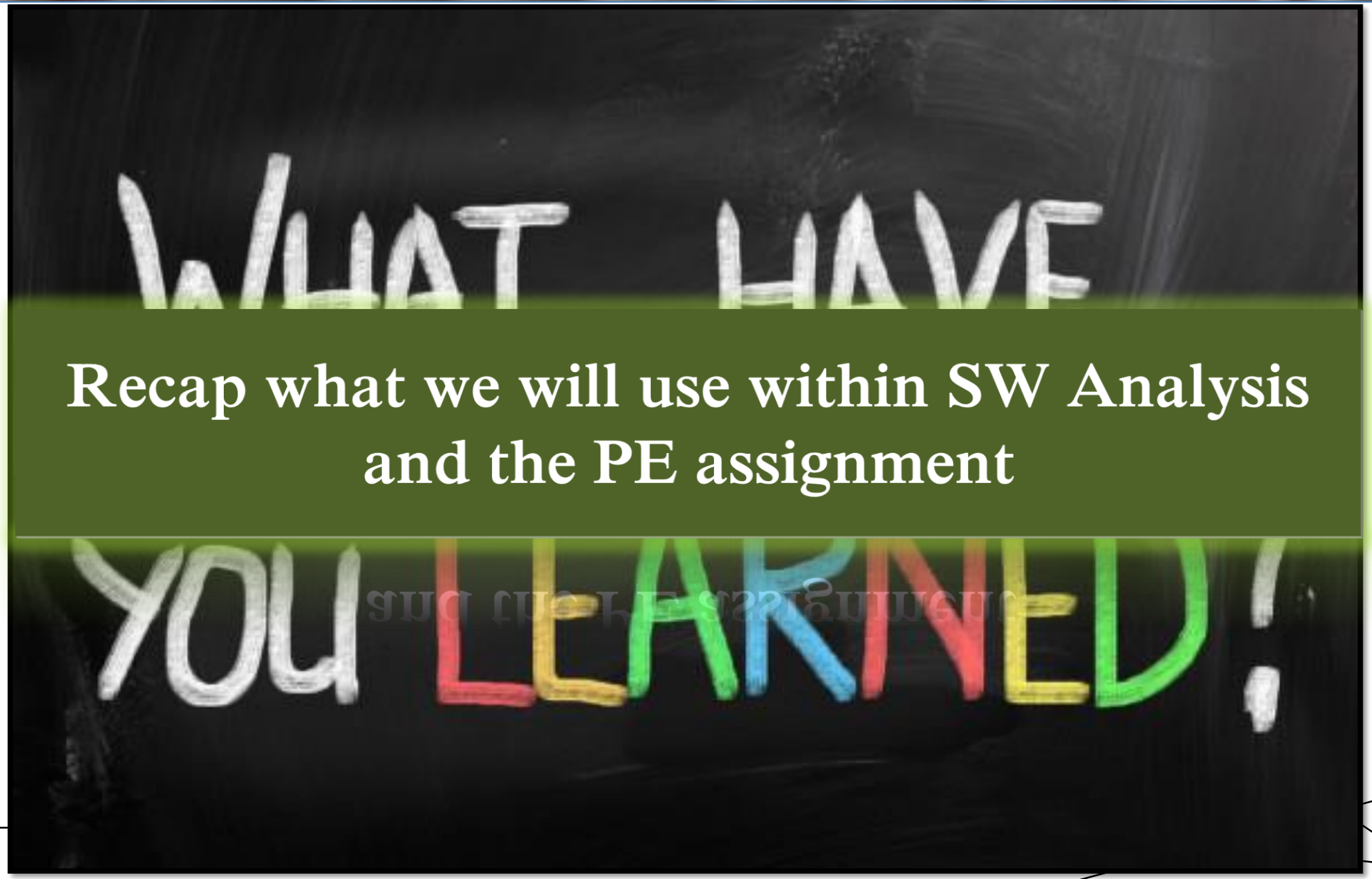
Complete requirement template (according to Rupp – Dutch version)



Dealing with transformations

Other useful templates of the working field

- The <stakeholder> shall be able to <capability> **URS**
  - The order clerk shall be able to raise an invoice
- The <product> shall be able to <action> <entity> **SRS**
  - The launcher shall be able to launch missiles
- The <product> shall provide <user> <function> <object> every <performance> <unit>
  - The coffee machine shall provide the user the ability to get a hot drink every 10 seconds



Recap what we will use within SW Analysis and the PE assignment

RECAP - User Story Template / Blueprint

As a <role>
I want <goal>
so that <benefit>

Acceptance criteria:
(Conditions of Satisfaction)
...
...

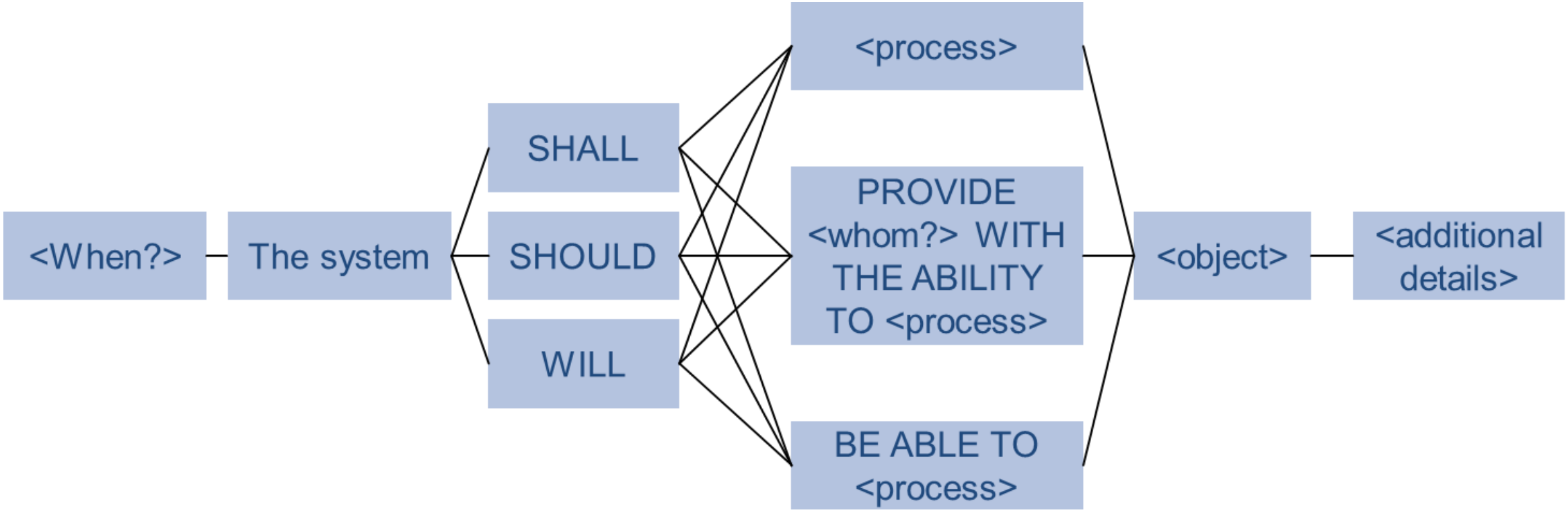
RECAP - User Story Examples

- Functional User story
  - As a registered user, I want to be able to customize my profile by uploading a profile picture so that my profile reflects my identity

- Non functional user story
  - As a system administrator, I want the data retrieval process to be swift and efficient (1000 records from the database in less than 3 seconds), so that I can quickly generate reports and provide timely information to end-users

RECAP – System Requirement Template / Blueprint

Complete requirement template (according to Rupp – English version)



RECAP – System Requirements Examples

- Functional System Requirement
  - If a user is registered, the system shall allow to personalize his profile by uploading a profile picture.

- Non functional System Requirement
  - When system administrators want to report on system behavior, the system shall ensure swift and efficient data retrieval (display 1000 records from the database in less than 3 seconds) to facilitate rapid report generation and timely information provision for system administrators

Recap IEEE 830 – SRS, standardized document structures & requirements classification

Other Reasons for Documentation

- Requirements ‘may’ have a long lifespan
- Accessibility for many persons
- Legal relevance / compliance
- We need a good overview
- Requirements may become very complex
- Stakeholder needs
  - Users (are my whishes / needs satisfied?)
  - Project Manager (consequences for budget/schedule)
  - System Engineers (translate to solution/architecture)
  - Testers (testability, traceability to test cases)
  - Software Engineers (translation to design and code)
  - Subcontractors (define work/contract – outsourcing!)
  - Change Management, Operation & Maintenance…

Two aspects

- Needs to be <u>readable</u>
  - The structure of the document, how it is **<u>organized</u>** and how the flow **<u>supports the reader</u>** to place individual requirements in context

- Needs to be <u>manageable</u>
  - Qualities of individual requirements, clarity and preciseness and how they are divided into **<u>single traceable items</u>**
  - MS-Word doesn't provide attributes, identifiers etc.

Perspectives on system to be developed

- Data perspective
  - Structure of data
  - Usage and dependency relationships in context
- Functional perspective
  - What information (data) from the system context is manipulated by the system and what data flows from the system to the system context
- Behavioral perspective
  - Documenting the system, embedded in the system context, in a state-oriented way

Writing the specification (cf. PowerPoint SRS)

- Three main sections
  - Introduction
  - Overall description
    - Constraints
  - Specific requirements
    - Functional requirements (grouped)
    - Quality requirements
- Therefore ➔ IEEE

Putting together what we have gathered so far

Why do we need a glossary?

BOW

- Homonyms
- Synonyms
- Common terms with a specific meaning in the context of the project
- Specific terms of the domain
- Technical terms
- Acronyms and abbreviations
- …
- Company specific and supplementary project specific

Writing requirements

- Functional requirements:
  - Describe a capability that the stakeholder needs or an action the product must take
    (what does the product have to do to complete this use case step?)
  - Use the form: “The product shall be able to do a specific thing (for a specific actor)”
- A single sentence with a single indicative verb - **<u>use simple direct language</u>**
- Don’t write a solution. Examine your requirements for unwanted technology

Guidelines

- Short and concise sentences and paragraphs
- One requirement per sentence (no compound requirements), no nesting

The 2 most important style rules!

- Consistent terminology
- Avoid generalizations, clear references
- Use ‘must’, ‘can’ and similar words carefully (‘shall’ is better)

Quality gateway

- Each individual requirement must pass through the quality gateway to be added to the specification – it's the only way in …
- Requirements must be …
  - Agreed
  - Ranked
  - Unambiguous
  - Valid and up-to-date
  - Correct
  - Consistent
  - Verifiable
  - Realizable
  - Traceable
  - Complete
  - Understandable
  - …

The rule set!

Rules and checklists

- See any defects with this requirement?

“The objective is to get higher adaptability using product Y”

Rules for quality requirements

Example

1. They should be unambiguously clear to the intended reader.
2. They shall specify a scale of measure to define the concept.
3. They shall break down complex concepts into a set of measurable concepts.
4. To define "relative" terms like "higher" they shall specify at least two points of reference on the defined scale.
5. They shall specify exactly when a quality level is to be available.
6. They shall not mix design ideas in the specification of objectives.
7. The process input or "source" (like contract, standard, marketing plan) of the requirement shall be given.

Rules and checklists

- See any defects with this requirement?

"The objective is to get higher adaptability using product Y"

No statement of exactly <u>when</u> the objective is to be met (5)

Ambiguous, unclear (1)

No 2 points of reference to define "higher"(4)

No SCALE (2)

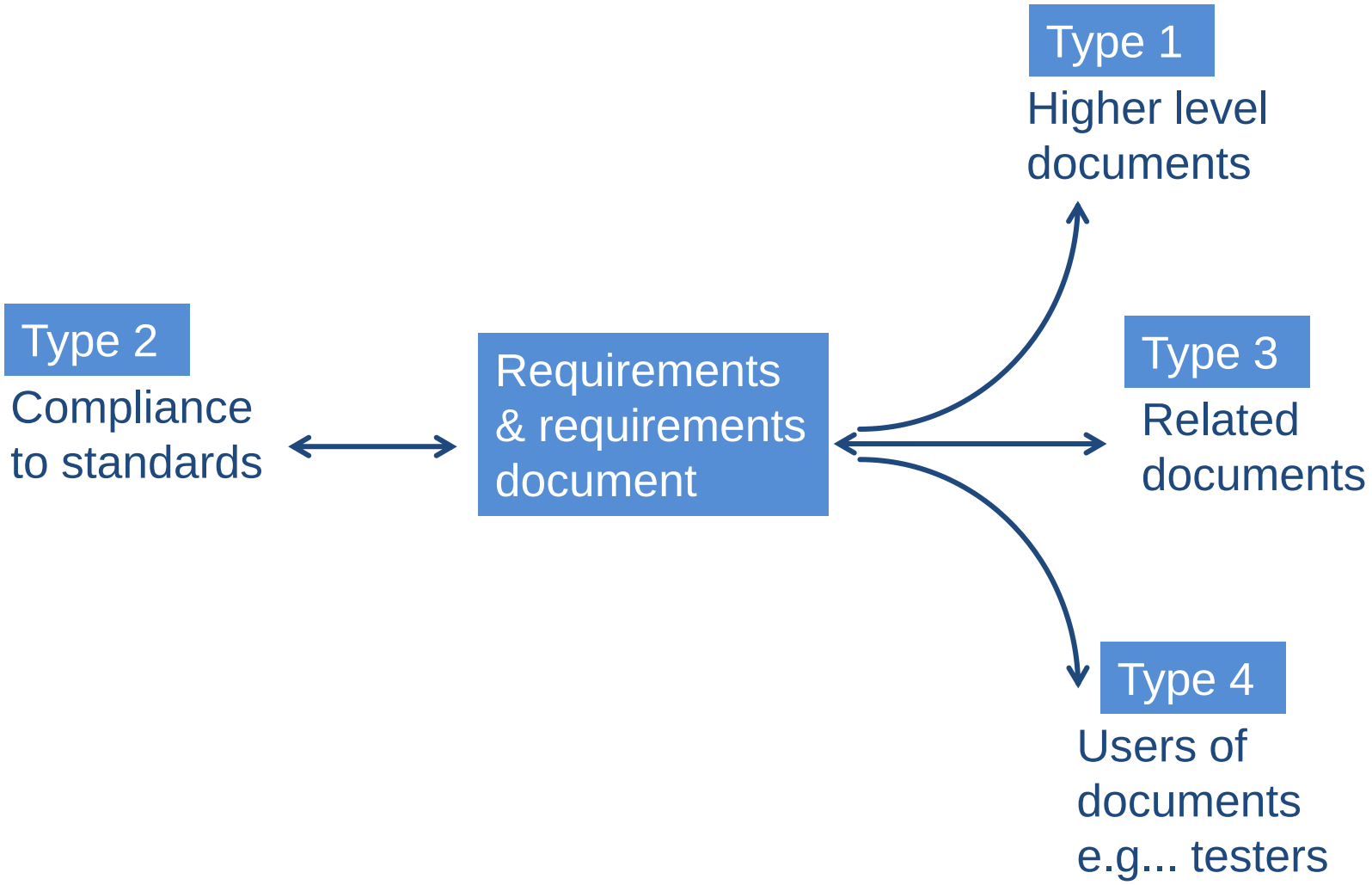
Complex concept not broken down (3)

Source not given (7)

A design idea is mixed into the objective (6)

Rules

Shall be company specific!



Fit criteria

Fit criteria (1)

Involve tester here!

- "Fit" means that a solution completely satisfies the defined requirement
- Need to attach a quantification to the requirement
  - Make it measurable
  - For use cases this is the post condition (outcome)
  - The quantification of the requirement is its fit criterion
- The fit criterion may quantify the behavior, the performance, or some other quality of the requirement
  - So more important for quality requirements
  - Functional requirements should already be testable

Fit criteria (2)

- Fit criteria apply to both functional and non-functional requirements
- !!! Analyze requirement description and determine requirement rationale to find the appropriate scale of measurement for fit criteria
- Make it possible to determine <u>whether</u> a <u>requirement</u> has been <u>satisfied</u> or not
- It is usually necessary <u>to negotiate</u> fit criteria with the <u>stakeholder</u>
  - E.g., 90% of the customers must be able to get the correct ticket from the product in no more than 25 seconds

Fit criteria (3) – Examples (1)

- Functional User Story
  - As a registered user, I want to be able to customize my profile by uploading a profile picture so that my profile reflects my identity.

  - Acceptance Criteria:
    1. The system should provide an option for users to upload a profile picture
    2. The uploaded picture should meet the specified file format (e.g., JPEG, PNG)
    3. The system should display a preview of the selected picture before confirmation
    4. Upon confirmation, the uploaded picture should be saved to the user's profile
    5. Users should be able to update or remove their profile picture at any time.

Fit criteria (3) – Examples (2)

- Non-functional User Story
  - As a system administrator, I want the data retrieval process to be swift and efficient (1000 records from the database in less than 3 seconds), so that I can quickly generate reports and provide timely information to end-users
  - Acceptance Criteria:
    1. The system should retrieve and display 1000 records from the database in less than 3 seconds.
    2. The system's response time for data retrieval shall not significantly increase, maintaining a maximum increase of 20% as the database grows.
    3. The system's response time shall meet specified benchmarks for consistent performance based on industry standards.

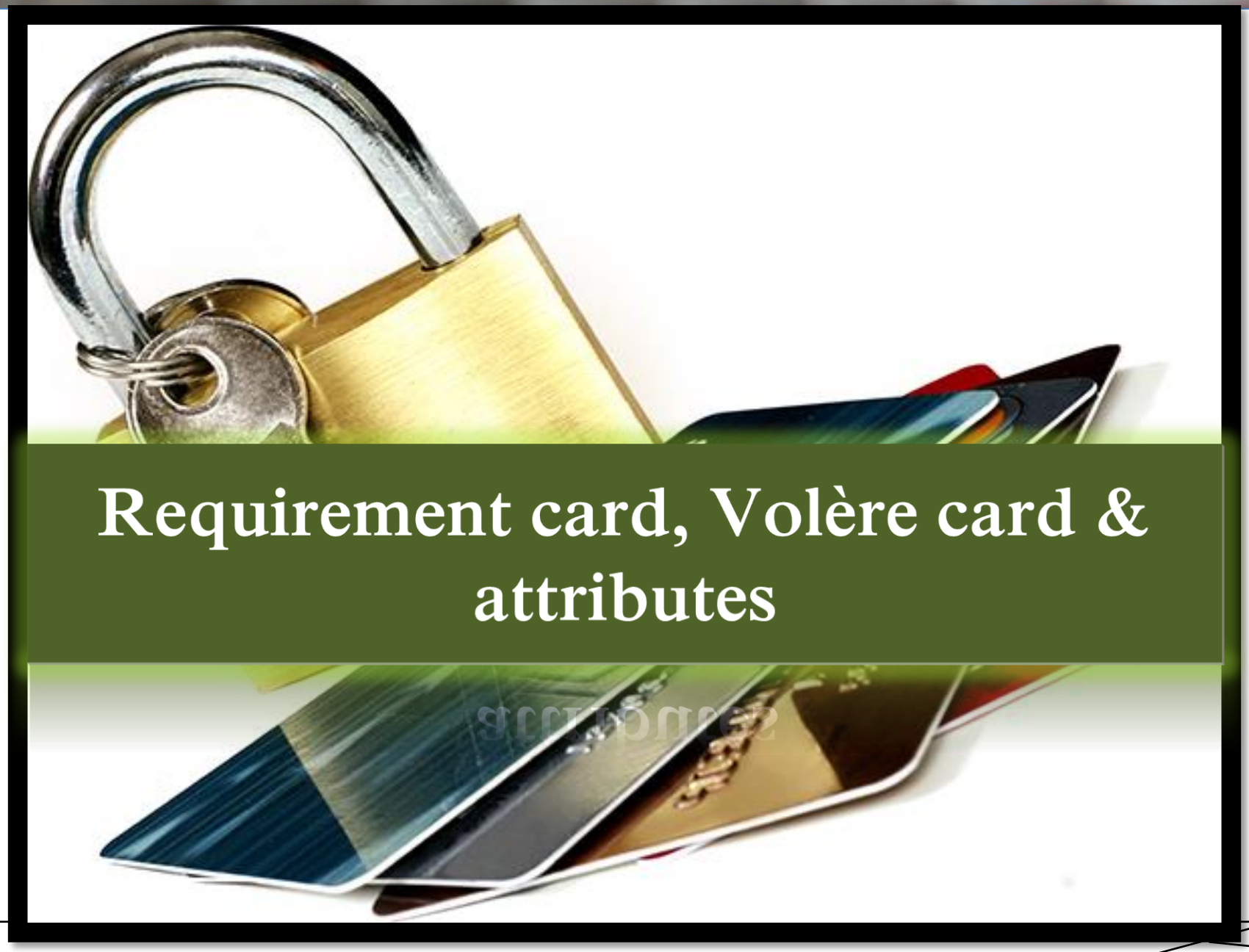
Fit criteria (3) – Examples (3)

- Functional System Requirement
  - When an instructor wants to add content, the system must give the instructor the ability to upload various types of educational content, including Word documents, Excel sheets, PowerPoint presentations, MP4 videos, and interactive quizzes

  - Acceptance Criteria:
    1. The system must successfully accept and process uploads of Word documents (.docx), Excel sheets (.xlsx), PowerPoint presentations (.pptx), MP4 videos, and interactive quiz files, without data corruption.
    2. Instructors should be able to upload Word documents, Excel sheets, and PowerPoint presentations up to 50 MB in size, and MP4 videos and interactive quizzes up to 500 MB, with the system preventing uploads exceeding these limits.

Fit criteria (3) – Examples (4)

- Non-functional System Requirement
  - When an instructor wants to add content, the system should facilitate uploads from various devices and operating systems commonly used by instructors, such as Windows, macOS, and common web browsers (Chrome, Firefox, Safari).
  - Acceptance Criteria:
    1. The system must support file uploads from Windows 10 (or the latest version) and macOS 10.14 (or the latest version), ensuring compatibility with the two major operating systems used by instructors.
    2. ile uploads should be fully functional on the latest versions of commonly used web browsers, including Google Chrome, Mozilla Firefox, and Apple Safari.
    3. The system shall support Microsoft Edge for users on Windows and Safari for users on macOS.



Requirement card, Volère card & attributes

Requirements card – Volère card

A lot of examples to be found on the net

Requirement #		***Priority***	
Requirement title			
Requirement Type		***Use Case***	
Description User story			
Description System req.			
Rationale			
Source			
Fit Criterion :			
Supporting material		***Annotation[1]***	

Requirements card – Volère card

Blue prints
- User story
- System Req.

Requirement #	Unique number of the requirement	*Priority*	Based on the words Shall/Must (=1), Should (=2), Will (=3)
Requirement title	Kind of summary indicating the requirement		
Requirement Type	PERFUMeS or business requirement	*Use Case*	Number of the linked use case in use case diagram and description
Description	Give a SMART description in natural language of the requirement. Describe the intend of the requirement. Use the BLUEPRINTS of a user story and a system requirement		
Rationale	The reason WHY this requirement needs to be developed		
Source	Stakeholder (group) of the requirement. For which user this requirement is defined?		
Fit Criterion :	Acceptance criterion defined in a quantified manner. To be used for acceptance testing		
Supporting material	Link to further material	*Annotation*[1]	Eventual remarks concerning the requirement

Requirements attributes (1)

- ID
  - To allow traceability
- Type
  - Allows req.'s to be sorted, grouping allows the requirements to be checked on completeness and for conflicts, e.g. by non-functional, by business process
- Use case number
  - For traceability and change control purposes
  - Again for grouping etc.
- Description
  - The intent of the requirement (may initially be ambiguous) - the stakeholders' whishes & needs

Requirements attributes (2)

- Rationale
  - Reason behind the requirement's existence. Helps to clarify and understand the requirement. (to find 'gold plating' req.'s)
- Source
  - Name of the person who raised the requirement.
- Priority
  - Measure of the business importance. For negotiation, but also for risk-based testing.
- Others …
  - Dependencies, supporting materials, annotation, history

Exercise – new clock radio (part 02)

- You have already found defects in the requirements for the new clock radio
- Assignment
  - As a team re-write both functional and quality requirements for the new alarm clock
  - Use the MS Word template (Volère) and fill out (at least) the following attributes:
    - Unique id, description (both user story and system requirement), rationale and when appropriate the fit criterion. Source = your name!
    - System level requirements, English language

Questions & answers